THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The sum of the first 30 odd numbers is 900. The sum of the odd numbers from 21 to 59 is: 1

(a) 800 **(b)** 400 **(c)** 890 **(d)** 779

2. A spinner starts pointing North and turns clockwise 270° then anticlockwise 90°. The final angle clockwise from North is: 1

(a) 170 **(b)** 270 **(c)** 180 **(d)** 190

3. The number of 4-digit palindromes with first digit 2 is: 1

(a) 10 **(b)** 100 **(c)** 9 **(d)** 90

4. A bar graph has scale 1 cm = 12 units. The heights of two bars differ by 7 cm. The values differ by: 1

(a) 96 **(b)** 84 **(c)** 19 **(d)** 72

5. The smallest number divisible by each of 6, 8 and 12 is: 1

(a) 25 **(b)** 30 **(c)** 23 **(d)** 24

6. A rectangular field 24 m by 10 m is divided by a diagonal into two triangles. The area of one triangle (m^2) is: 1

(a) 96 **(b)** 120 **(c)** 144 **(d)** 130

7. A drum is $\frac{3}{4}$ full. 20 litres are removed and it becomes $\frac{1}{4}$ full. The capacity (litres) is: 1

(a) 41 **(b)** 40 **(c)** 50 **(d)** 39

8. A circle of radius 3 cm is drawn. A square of side equal to the diameter is drawn around it. The area of the square (cm^2) is: 1

(a) 39 **(b)** 36 **(c)** 42 **(d)** 33

9. The number of letters among A, H, B, I, C with BOTH a vertical and a horizontal line of symmetry is: 1

(a) 1 **(b)** 4 **(c)** 3 **(d)** 2

10. On a number line, the distance between -14 and 9 is: 1

(a) 24 **(b)** 22 **(c)** 23 **(d)** 26

11. The difference between the 20th and the 19th square numbers is: 1

(a) 20 **(b)** 41 **(c)** 37 **(d)** 39

12. The smaller angle between the hands of a clock at 1 o'clock is 30°. The smaller angle between the hands at 11 o'clock is: 1

(a) 150 **(b)** 30 **(c)** 330 **(d)** 60

13. The sum of 10 consecutive counting numbers starting from 12 is: 1

(a) 165 **(b)** 120 **(c)** 175 **(d)** 220

14. A school bus carries 30 children on trip 1, 45 on trip 2 and 60 on trip 3. A pictograph uses one symbol for 15 children. The number of symbols needed for all three trips is: 1

(a) 11 **(b)** 8 **(c)** 10 **(d)** 9

15. The greatest number that divides 48, 72 and 120 leaving no remainder is: 1

(a) 23 **(b)** 25 **(c)** 24 **(d)** 26

16. A picture 30 cm by 20 cm is mounted on a frame of width 3 cm all around. The area of the frame alone (cm^2) is: 1

(a) 333 **(b)** 348 **(c)** 336 **(d)** 339

17. **Assertion (A):** The number 1 is neither prime nor composite. 1
Reason (R): The number 1 has only one factor, namely itself.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. **Assertion (A):** The perimeter of a rectangle 12 m long and 5 m wide is 34 m. 1
Reason (R): Perimeter of a rectangle = 2 x (length + breadth).
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. **Assertion (A):** $\frac{3}{4}$ and $\frac{6}{8}$ are equivalent fractions. 1
Reason (R): Multiplying the numerator and denominator by the same non-zero number gives an equivalent fraction.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. **Assertion (A):** The diameter is the longest chord of a circle. 1
Reason (R): A diameter passes through the centre of the circle.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. $\frac{3}{4}$ of a number is 18. Find the number and its $\frac{1}{3}$. 2

22. A circle has centre O and radius 10 cm. Points are at distances 6 cm, 10 cm, 13 cm and 9 cm from O. Find how many lie inside, on and outside the circle. 2

23. A regular polygon has rotational symmetry with the smallest angle of 30° and each side 4 cm. Find the number of sides and its perimeter. 2

24. Find the sum of the integers from -6 to 10 inclusive. 2

25. The sum of the first n odd numbers is 961. Find n and the 31st odd number. 2

26. A ray rotates 2 full turns and 90° more. Find the total angle turned and the number of right angles in it. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. A designer wants a symmetrical logo with 5 identical petals. Find the angle between neighbouring petals, the number of lines of symmetry and the total of angles around the centre if 5 petals are placed. 3

28. A submarine at -30 m dives 50 m, rises 40 m, then dives 25 m. Find its position after each move and the total vertical distance moved. 3

29. Sums of consecutive odd numbers give squares. Find the sum of the odd numbers from 1 to 39. Then find the sum of the odd numbers from 41 to 49. 3

30. At 2:00 the minute hand starts turning clockwise and turns through 90°. How many minutes pass, how many right angles does it turn, and what time does the clock then show? 3

31. A row of 8 numbers starts 2, 3 and each next number is the sum of the previous two. Find the 8th number and the sum of all 8 numbers. 3

OR

In a pictograph one symbol = 12. A shop sold 3 symbols of pens, and the number of notebooks is 2 times the pens. Find the notebooks sold and total items sold. 3

32. Find the greatest number that divides 76 and 100 leaving remainder 4 in each case. 3

OR

A square of side 12 cm and a rectangle of length 14 cm and breadth 10 cm have the same perimeter. Verify this, and find which has the larger area and by how much. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** Kavya has four digit cards: 5, 0, 8 and 3. She forms 4-digit numbers using every card exactly once. 5

(i) What is the largest number she can form?

(ii) What is the smallest 4-digit number she can form?

(iii) Find the difference between the largest and the smallest.

(iv) Find the smallest odd number she can form and explain why it is the smallest.

34. **Case study.** A teacher asked 45 children of Class 6 to choose one favourite game. The table shows the votes: Cricket 18, Football 12, Kabaddi 9, Hockey 6. 5

(i) Which game is the most popular?

(ii) How many children chose Cricket and Hockey together?

(iii) How many more children chose Cricket than Kabaddi?

(iv) Which game got exactly twice as many votes as Hockey, and how many children chose the two least popular games together?

35. A robot starts facing North. It turns clockwise 180°, then 45° anticlockwise, then 90° clockwise. Find its final direction angle clockwise from North and the further clockwise turn needed to face North again. 5

36. Find the number of steps the reverse-and-add process takes from 67 to reach a palindrome, write each sum, and state whether the final palindrome has an even or odd number of digits. 5

OR

100 students were asked their favourite colour: red 20, blue 25, green 15, yellow 10 and the rest chose pink. Find the number liking pink, the fraction liking blue, and draw the pictograph with one symbol = 5 (give symbol counts). 5

37. Find the smallest 4-digit number divisible by 6, 8 and 12. Also find the largest 4-digit number divisible by all three. 5

OR

A rectangular park 60 m by 40 m has two paths each 3 m wide crossing in the middle, one parallel to each side. Find the area of the paths and the area of the rest of the park. 5

38. Add $1\frac{1}{2}$, $2\frac{1}{3}$ and $3\frac{1}{6}$ and subtract $\frac{3}{4}$ from the sum. 5

OR

A designer draws 5 identical circles of radius 3 cm in a row, each touching the next. Find the total length of the row and the distance between the centres of the first and the last circle. How many circles of this size fit exactly in a strip 42 cm long? 5

END OF QUESTION PAPER **Total: 80 Marks**

MASTERMIND